

Bankruptcy prediction using optimal ensemble models under balanced and imbalanced data

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Abstract

This study explores the performance of gradient boosting methods in bankruptcy prediction for a highly imbalanced dataset. We developed different heterogeneous ensemble models based on three popular gradient boosting methods—XGBoost, LightGBM, and CatBoost. Our ensemble models were optimized using the cross-validation method and the results of the hold-out test sets showed that the optimized ensemble models not only outperform their base learners, but also improve the state-of-the-art benchmark results on the same dataset. Interestingly, we observed that the data oversampling technique that is commonly used to address the class imbalance issue had an adverse impact on our ensemble models' performance. This indicates that our models are robust to the imbalanced dataset problem that typically degrades the classification performance of machine learning models.

KEYWORDS

bankruptcy prediction, gradient boosting methods, imbalanced dataset, optimal ensemble models

1 | INTRODUCTION

Bankruptcy is, in general, the consequence of financial distress, a situation in which a firm is not able to fulfil its obligations (Zoričák et al., 2020). Bankruptcy has significant costs for managers, shareholders, employees, and more (Veganzones & Séverin, 2018). Bankruptcy can have various causes, including legal, economic, financial, strategic, organizational, and managerial factors (Jabeur et al., 2021). Predicting the likelihood of bankruptcy for a company is important because it helps to identify potential financial distress early, take proactive actions to reduce risk, save costs, and aid in financial planning for the company before the company reaches the critical stage. On the other hand, accurate prediction of bankruptcy allows banks and financial institutions to identify and manage their risk exposure. If they can predict that a company is likely to go bankrupt, they can reduce their exposure to that company's securities or loans, minimizing their potential losses. Bankruptcy prediction is a binary classification task where the goal is to predict whether a company will go bankrupt in the future or not. In this regard, various machine learning models and features selection techniques have been used to accurately classify bankrupt companies (Lahmiri, 2016; Lahmiri & Bekiros, 2019). Besides, in the real-world scenario, bankruptcy is a rare occurrence for most companies, and this presents a challenge for the classification task at hand.

The literature offers various strategies to address the challenge of achieving accurate predictions in such scenarios. One of the most widely used techniques is sampling (e.g., in Veganzones & Séverin, 2018; Papík & Papíková, 2023) which involves either adding (oversampling the minority class) or removing (undersampling the majority class) samples from the existing dataset to achieve a balanced bankruptcy dataset. The Synthetic Minority Oversampling Technique (SMOTE) proposed by Chawla et al. (2002) is the most popular one. Various versions of SMOTE have

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been proposed by others. We refer the reader to the book by Fernández et al. (2018) which thoroughly reviews the extensions of SMOTE method.

Another solution for addressing the issue of an imbalanced dataset is the utilization of ensemble models (Lahmiri et al., 2021; Lahmiri, Bekiros, et al., 2020). Indeed, ensemble learning or ensemble modelling is defined as combination of predictions from different machine learning models to make the final prediction and it has been proven to increase the performance of the individual models. As highlighted by Ganaie et al. (2022), the success of ensemble methods is attributed to the following three reasons:

1. Bias-variance decomposition: Bias is the difference between the model's expected prediction and the actual value, while variance quantifies the variability in the model's predictions across different training sets, measuring its sensitivity to minor changes in the training data. Ensemble methods combine predictions from multiple models, each with its own bias, thereby mitigating the overall prediction bias. Simultaneously, this aggregation minimizes prediction variance.
2. Statistical, computational, and representational aspects: By combining different classifiers, the ensemble models can reduce the overfitting problem and overcomes the local optima search. In addition, more complex patterns can be identified by ensemble methods that individual models might overlook.
3. Diversity: Ensemble methods add diversity to the predicted outputs by generating multiple datasets from the original dataset, contributing to improved model generalization.

The contributions of this article are summarized as follows:

- We designed various heterogenous ensemble models using three gradient boosting classifiers: XGBoost, LightGBM, and CatBoost. To the best of our knowledge, the application of these classifiers in bankruptcy or financial distress prediction is rare in existing literature, and the ensemble of them has never been explored by other researchers in this domain.
- We showed the superiority of the ensemble models over their individual base learners and also other benchmark results from existing literature, obtained on the same dataset.
- In contrast to prior studies employing either cross-validation or hold-out test set strategies for model evaluation, we utilized cross-validation to determine optimal weights for base classifiers in our ensemble models. We then evaluated these optimized models on hold-out test sets, showing that they could accurately predict bankruptcy for new data.
- We applied the commonly used data oversampling technique of SMOTE to our imbalanced dataset and we showed that the application of these technique is not always beneficial though it has been shown as an effective solution for the classification problems in highly imbalanced datasets.

The rest of this article is organized as follows: Section 2 reviews recent studies in bankruptcy prediction followed by a brief review on application of ensemble learning methods in different domains. Section 3 explains the theoretical background of our research methodology. Section 4 describes the data and results. In Section 5, we discuss the results. Finally, the conclusion and future research ideas are given in Section 6.

2 | LITERATURE REVIEW

The following section reviews most relevant recent studies in the domain of bankruptcy prediction.

2.1 | Bankruptcy prediction

Considering the highly imbalanced nature of bankruptcy datasets, Veganzones and Séverin (2018) addressed two fundamental questions: Which imbalanced class distribution disturbs a model's predictive performance? And what is the improvement capacity of treatment techniques? Four sampling techniques were employed by the authors, namely, random oversampling, random undersampling, the synthetic minority oversampling (SMOTE), and EasyEnsemble. The classifiers in this study were linear discriminant analysis (LDA), logistic regression (LR), neural network (NN), support vector machine (SVM), and random forest (RF). As a result, they found that when a dataset has an imbalanced proportion of 4:1 (data that contains 80% nonbankrupt and 20% bankrupt firms) or higher, the models' ability to predict bankruptcy is jeopardized. They also found that SVM models are least influenced by imbalanced proportions.

Zoričák et al. (2020) employed the one-class classification methods using only samples from the majority class to train their models. The anomalous samples in the test dataset were labelled as bankrupt. Using highly imbalanced bankruptcy datasets in the construction and manufacturing sectors, they trained and compared three one-class classifiers, namely, One-Class SVM (OCSVM) (Schölkopf et al., 2001), Isolation

Forest (IF) (Liu et al., 2008) and Least-Squares Anomaly Detection (LSAD) (Quinn & Sugiyama, 2014) with support vector machines (SVMs). The LSAD model outperformed others with an AUC value of 91.83%. In addition, they found that the highest prediction performance was achieved when only data from the previous year of the evaluation year is used for prediction.

Employing 11 classifiers and using 6 sampling techniques, Garcia (2022) showed that training 7 out of 11 classifiers on balanced datasets significantly improved the recall measure. The author claimed that the cost of bankruptcy is high, and the recall is the most important performance measure for bankruptcy predictions. In addition, the RF ensemble method, trained using any synthetically balanced dataset was found to be the best-performing classifier across all evaluation metrics.

The most popular and commonly used ensemble methods in bankruptcy prediction are bagging (Breiman, 1996) and boosting (Freund, 1995). Reviewing the recent studies in bankruptcy prediction, we realized that the gradient boosting algorithms such as XGBoost (Chen & Guestrin, 2016) and LightGBM (Ke et al., 2017) have shown superiority over other machine learning models in addressing the challenges of bankruptcy prediction.

XGBoost was first used in the work of Zięba et al. (2016), where they compared the prediction performance of different classifiers for predicting bankruptcy in a highly imbalanced dataset. The authors introduced a new approach called 'synthetic features', using arithmetic operations (addition, subtraction, multiplication, and division) on econometric measures to generate more complex features. They defined five classification periods where they predict the bankruptcy status of companies after 1–5 years. They noticed that the XGBoost model significantly outperformed other models across all classification periods. They even improved the results of the XGBoost model by further training the last base learner of this model on synthetic features.

Another boosting method that has recently demonstrated high performance in predicting bankruptcy is the CatBoost model (Prokhorenkova et al., 2018) which handles categorical features in a more efficient way than other popular gradient boosting algorithms such as XGBoost and LightGBM. The first article that employed CatBoost in the bankruptcy and corporate failure prediction problem, was Jabeur et al. (2021). In their study, which involved training and testing nine classifiers on corporate failure data from France, CatBoost outperformed other classifiers such as XGBoost, GBM, Deep Neural Networks (DNNs), SVM, etc., in terms of both accuracy and Area Under the ROC Curve (AUC) measures. Particularly, at a very short horizon before corporate failure, that is, 1 year before the failure year, this model could achieve an AUC value of 99.4%. The imbalanced dataset issue was not present in this study.

More recently, Papík and Papíková (2023) employed three gradient boosting methods, namely, CatBoost, LightGBM, and XGBoost to analyse the impact of crisis periods on bankruptcy prediction models and their performance. Considering two non-crisis periods and one crisis period, they showed that the performance of prediction models was significantly weaker during crisis periods compared to non-crisis periods.

Regarding the use of multiple classifiers in an ensemble way, most of the previous studies in bankruptcy prediction domain employed traditional machine learning models such as SVM, LR, MLP, and Decision Trees as their base learners. The use of gradient boosting-based models as the base learners of ensemble models was found in the work of Du et al. (2020) where the authors implemented an ensemble of Gradient Boosted Decision Trees (GBDT) (Friedman, 2001) and XGBoost to predict financial distress of Chinese companies. In fact, they used the concept of ensemble learning throughout the entire process, from feature selection to model construction. The ensemble of five feature selection methods identified the most important features. Then, with combining clustering-based under-sampling (CUS) proposed by Rayhan et al. (2017) with GBDT, an ensemble model was built based on CUS-GBDT and XGBoost. Their proposed heterogeneous ensemble obtained the best performance with an AUC of 97.75%, which is superior to that of the component learner CUS-GBDT (AUC of 96.45%) and the XGBoost (AUC of 93.22%). Table 1 summarizes the recent studies in bankruptcy prediction.

Our study is founded upon the works of Zięba et al. (2016) and Papík and Papíková (2023). Specifically, we build different heterogeneous ensemble models based on the weighted average combination of the three classifiers employed by Papík and Papíková (2023) and will use the polish companies' dataset from Zięba et al. (2016) to explore the effectiveness of these ensemble models. Our study is different from the study of Zięba et al. (2016) in several ways: (1) We demonstrate the predictive capabilities of two additional gradient boosting models, namely, the CatBoost and LightGBM, alongside XGBoost to predict bankruptcy in polish companies' dataset. (2) We introduce new ensemble methods that outperform the results reported by Zięba et al. (2016). (3) We explore the effectiveness of oversampling methods and show that our models perform better on the original (imbalanced) data compared to datasets that have undergone oversampling. This underscores the robustness of our models in addressing imbalanced data issues.

2.2 | Application of ensemble methods in other domains

Ensemble learning methods have been successfully employed in various domains such as health care (Cabria & Gondra, 2017; Gifani et al., 2021; Zhou et al., 2021), text classification (Kilimci & Akyokus, 2018; Mohammed & Kora, 2022), marketing (Lessmann et al., 2021; Martínez et al., 2020), speech recognition (Deng & Platt, 2014; Pravin et al., 2023), and numerous other fields.

In the healthcare domain, Cabria and Gondra (2017) evaluated the performance of different methods, including an ensemble of individual segmentation methods, on the publicly available brain tumour Image benchmark dataset. They showed that the ensemble model improved the overall

TABLE 1 Recent studies in bankruptcy prediction.

References	Data (period)	Features	Class distributions	Imbalanced data treatment	Models	Evaluation method	Performance metrics	Outcome
Veganzones and Séverin (2018)	1500 French firms in different sectors: service, construction, retail, others.	50 financial ratios calculated based on balance sheets and income statements	5% bankrupt	Four sampling techniques	LDA LR, NN, SVM, and RF	Hold-out test set (20%)	Sensitivity, specificity, G-mean, and AUC	The SVM and RF methods significantly outperform the rest of the methods
Zoričák et al. (2020)	Business entities operating in the Slovak republic (2010–2016)	20 financial attributes derived from the company's annual report 3 years prior to the year of evaluation	2% bankrupt	One-class classification methods	OCSVM, IF, LSAD, SVM	Hold-out test set (20%)	G-mean and AUC	The highest prediction scores were obtained by LSAD
Zięba et al. (2016)	Polish companies (2007–2013 for bankrupt and 2000–2012 for still operating companies)	64 financial indicators based on financial statements	Between 3% to 7% bankrupt for different classification periods	N.A.	16 classifiers including LDA, SVM, RF, XGB, etc.	10-folds CV	AUC	The ensemble extension of the XGB model outperformed all classifiers
Jabeur et al. (2021)	Financial data for French firms (2014–2016)	18 ratios based on financial reports for 3 years before 2017	N.A.	N.A.	8 classifiers including DA, LR, SVM, RF, GBM, XGBoost and CatBoost	Hold-out test set (30%)	Accuracy and AUC	CatBoost achieved an AUC value of 99.4%.
Garcia (2022)	1824 publicly traded firms in the United States (2010 to 2018)	3 financial ratios along with 2 market variables	2.6% bankrupt	Six synthetic balancing techniques	12 classifiers including LR, LDA, SVM, XGBoost, RF, RF ensemble, etc.	Hold-out test set (30%)	Precision, Recall, Specificity, G-mean, AUC, F-measure	The best classifier was the RF ensemble trained using any synthetically balanced dataset
Papik and Papiková (2023)	More than 90,000 SMEs operating in Slovakia (2015 to 2019)	Six qualitative indicators and 46 financial ratios for the period of one, two and 3 years prior to actual bankruptcy	4%–5% bankrupt	Resampling of the training samples by SMOTE Technique	CatBoost, LightGBM, XGBoost	10-folds CV	Accuracy, Sensitivity, Specificity, AUC	The performance of prediction models was significantly weaker during crisis periods (on average, 4.1% less)
Du et al. (2020)	3670 listed companies in China's main board market in the 2018	21 selected features by 5 feature selection methods from the original 89 features	4% bankrupt	Combined CUS with the GBDT	Ensemble of GBDT and XGBoost	Hold-out test set (30%)	AUC, Accuracy, PPV	The proposed heterogeneous ensemble obtained the best performance

segmentation performance. One recent application of ensemble models can be seen in COVID-19 detection based on Computed Tomography (CT) scans. As an instance, Gifani et al. (2021) built an ensemble method based on majority voting of the outputs from different pre-trained Convolutional Neural Networks (CNN) to improve the recognition performance of the individual learners. In a similar application, Zhou et al. (2021) built an ensemble classifier using the relative majority vote algorithm to detect COVID-19 cases from CT scans. Their results showed that the overall classification performance of the ensemble model was better than that of the component classifiers, which included state-of-the-art CNN models such as AlexNet (Krizhevsky et al., 2012), GoogleNet (Szegedy et al., 2015), and ResNet (He et al., 2016).

In the field of text classification, Kilimci and Akyokus (2018) analysed the classification accuracy of different document representations by employing an ensemble of classifiers on eight different datasets. Their results show that the heterogeneous ensembles together with deep learning methods and word embeddings, enhances the classification performance of texts. Recently, Mohammed and Kora (2022) combined convolutional neural network (CNN), long short-term memory (LSTM), and bidirectional LSTM (BiLSTM) models to create an ensemble deep learning framework for text classification. Through multiple experiments on six public benchmark datasets, they concluded that their proposed ensemble method significantly improved the classification accuracy of the baseline deep learning models.

The application of ensemble models can be seen in the marketing domain as well. For example, Lahmiri (2017a) designed an ensemble neural network system to enhance the prediction of telemarketing outcomes and assist the marketing management team in the banking industry. The proposed system was composed of subsystems where each one is trained with a specific category of marketing information. The empirical results showed that the proposed system outperformed its individual components and the baseline model. Martínez et al. (2020) proposed a heterogeneous ensemble model that integrated multiple data sources for predicting customer purchase behaviour in the e-commerce industry. Lessmann et al. (2021) proposed a framework that included a feature selection module that identifies the most important customer characteristics, a clustering module that groups customers based on their similarities, and an ensemble module that combined multiple predictive models to generate targeted marketing recommendations. Their results showed that the framework outperformed several baseline methods in terms of profitability, and it could help companies make more effective marketing decisions and increase their profits by targeting the right customers with the right offers.

In the field of speech recognition, Deng and Platt (2014) combined multiple neural networks to create an ensemble model that outperformed individual models. The authors also investigated the impact of different ensemble strategies and found that the majority voting approach performs best. In a more recent study, Pravin et al. (2023) proposed two deep ensemble models, namely the Deep Cascaded Ensemble (DCE) and Deep Parallel Ensemble (DPE) for automatic speech emotion classification. The proposed DCE and the DPE were evaluated on various performance metrics and the DCE was observed to have better classification accuracy than the base learners.

In the field of financial forecasting, Lahmiri (2014a) designed an ensemble predictive system based on different artificial neural networks to predict NASDAQ price returns. It was found that ensemble models based on advanced NN architectures such as time-delay neural network, nonlinear auto-regressive with exogenous inputs, and radial basis neural network all achieve lower forecasting errors than traditional feed-forward back-propagation neural network ensemble system. In addition, the radial basis neural network ensemble outperformed all other ensembles of neural networks. Lahmiri (2014b) designed a radial basis function neural network (RBFNN) ensemble system (ES) to predict the Casablanca Stock Exchange price returns based on its microstructure modelling. The forecasting results showed that the ensemble system outperformed each of its components and the traditional ARMA model. Lahmiri and Boukadoum (2015a) proposed an ensemble system for stock market volatility which is composed of three different models that hybridize the exponential generalized autoregressive conditional heteroscedasticity (EGARCH) process and the artificial neural network trained with the backpropagation algorithm (BPNN) to forecast stock market volatility under normal, *t*-Student, and generalized error distribution assumption separately. The experimental results showed that proposed ensemble model used to capture normality, skewness, and kurtosis in data is more accurate than the individual EGARCH-BPNN models in forecasting the S&P 500 intra-day volatility. Lahmiri and Boukadoum (2015b) proposed a new ensemble system for stock market returns prediction where continuous wavelet transform (CWT) is used to analyse return series and backpropagation neural networks (BPNN) for learning CWT-based coefficients, and particle swarm optimization was considered for finding optimal weights and biases of each BPNN. To model symmetry and asymmetry in the data, three wavelet functions with different shapes were employed. The proposed ensemble system was tested on the Hang Seng, KOSPI, and Taiwan stock market data. The experimental results showed that the proposed ensemble system outperformed the individual CWT-ANN models each with different wavelet function. Furthermore, the proposed ensemble system outperformed the conventional autoregressive moving average process. Lahmiri (2017b) presented a variational mode decomposition (VMD) based generalized regression neural network (GRNN) ensemble learning model to predict California electricity and Brent crude oil prices. The particle swarm optimization method was employed to tune each GRNN initial weight within the ensemble system. The experimental results showed that the VMD-based ensemble outperformed the empirical mode decomposition (EMD) based ensemble forecasting system and the auto-regressive moving average model on both energy markets. Lahmiri (2018) presented an approach for analysing stock market price behaviour based on different categories of technical analysis metrics and a multiple predictive system. The proposed predictive system is based on an ensemble of neural networks coupled with particle swarm intelligence for parameter optimization where each single artificial neural network is trained with a specific class of technical analysis indicators. The experimental results on three international stock market indices and three individual stocks showed that the proposed ensemble-based technical indicators fusion predictive system meaningfully improves the forecasting accuracy in comparison with single neural network, neural network trained with index-level lagged values

and neural network trained with stationary wavelet transform details and approximation coefficients. Lahmiri, Saade, et al. (2020) proposed an ensemble model to forecast Bitcoin trading volume using radial basis function neural networks (RBFNN) and generalized regression neural networks (GRNN). They were adopted to respectively capture local and general patterns in Bitcoin trading volume data. Finally, the feedforward artificial neural network (FFNN) was implemented to aggregate Bitcoin final trading volume forecasts. The proposed ensemble predictive system reduced the forecasting errors by 18.81% and 62.86% when compared to its components RBFNN and GRNN, respectively. In addition, the ensemble system reduced the forecasting error by 90.49% when compared to a single FFNN used as a basic reference model.

To learn more about the different strategies and applications of ensemble learning, please refer to the work of Mohammed and Kora (2023) and Ganaie et al. (2022) which give comprehensive reviews of ensemble methods and their applications in various domains.

3 | METHODOLOGY

3.1 | Gradient boosting classifiers

The concept of gradient boosting algorithm was introduced by Friedman (2001) for the regression task but the same approach can be applied to classification problems as well. The algorithm involves iteratively adding new weak models to the ensemble model where each model is trained to minimize the residual errors of the previous model, improving the overall performance of the model. The final output is a weighted sum of basis functions. In the context of the classification, the algorithm begins with an initial model, often a decision tree, which makes predictions. The key idea is to iteratively improve the model's performance by focusing on the instances it misclassifies. A new weak classifier, typically a decision tree, is then trained to address these misclassifications. The ensemble, representing a weighted sum of the basis functions from all classifiers, is updated with the new weak classifier. This process is repeated for multiple iterations, refining the model with each step. The final result is an ensemble of weak classifiers that collectively offer improved and accurate predictions for classification tasks. Since the birth of this method, several boosting methods employing trees as weak learners have been introduced. In this study, three of these methods are utilized, and they are explained below.

3.1.1 | Extreme gradient boosting (XGBoost)

One popular implementation of gradient boosting for both classification and regression tasks is in Extreme Gradient Boosting (XGBoost) model that was developed by Chen and Guestrin (2016). XGBoost uses several optimization techniques such as parallel processing which reduces the computation time. For a given data set with n instances, the regularized objective function of the XGBoost model at iteration t is defined as following:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n l\left(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)\right) + \Omega(f_t), \quad (1)$$

where l is a differentiable convex loss function that measures the difference between the prediction $\hat{y}_i$ and the target y_i for instance i at previous iteration, that is, $t-1$. f_t and $\Omega(f_t)$ are the added decision tree and regularization term at iteration t , respectively. The regularization term penalizes the complexity of the model to prevent overfitting. The XGBoost model is formed with a single decision tree with one leaf node and by adding and training the f_t on residuals at t -th iteration, the weights of the previous trees in the model are updated so that the objective function is minimized. This process is repeated until a stopping criterion is met.

3.1.2 | Light gradient boosting machine (LightGBM)

Light Gradient Boosting Machine (LightGBM) (Ke et al., 2017) uses two techniques namely, Gradient-based One Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to reduce the training time and memory usage. The idea behind GOSS technique is to update the weights of decision trees based on samples that have larger gradients and consequently, more contribution to gradient boosting process. Then, the remaining samples are subsampled randomly for the next iteration. This randomly dropping the samples with small gradients in each iteration of gradient boosting algorithm speeds up the training process of the LightGBM model. For high-dimensional data, features are mutually exclusive meaning that they come with many zero values, and they do not take nonzero values simultaneously. The EFB technique bundles the mutually exclusive features with similar values and computes one single feature for each group by calculating their average. These new features are used in the training of the

model. This reduces the number of features that need to be evaluated in each iteration of gradient boosting algorithm and hence, reduces the training time.

The objective function of LightGBM for a classification task involves maximizing the log-likelihood of the observed labels based on the model's predictions. The formulation for the objective function at the t -th iteration can be expressed as:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \log \left(1 + \exp \left(-2y_i \hat{y}_i^{(t-1)} \right) \right) + \sum_{i=1}^n \Omega \left(f_i^{(t)} \right), \quad (2)$$

where: n represents the number of instances in the dataset. y_i denotes the true label of the i -th instance. $\hat{y}_i^{(t-1)}$ represents the model's prediction for the i -th instance at the $(t-1)$ -th iteration. $f_i^{(t)}$ denotes the i -th tree added to the model at the t -th iteration.

The first term in the objective function corresponds to the log-likelihood loss, capturing the likelihood of the observed labels given the model's predictions. The second term, $\Omega \left(f_i^{(t)} \right)$ is the regularization term that penalizes the complexity of the added tree to prevent overfitting.

3.1.3 | Categorical boosting (CatBoost)

CatBoost (Prokhorenkova et al., 2018) is similar to other gradient boosting algorithms—XGBoost and LightGBM—but it has some advantages over these two methods. The traditional gradient boosting algorithm tries all possible values of a feature in a node of the tree to find the best splitting point that results in the largest reduction of the loss function. The ordered boosting algorithm used in the CatBoost model considers only a pre-defined subset of threshold values that has been ordered for each feature during data splitting at each node. Reducing the number of possible split points not only improves the speed of the gradient boosting algorithm at each iteration, but it also reduces the memory usage as well.

Another advantage of the CatBoost model is the way that it handles categorical features. Using a technique called Ordered Target Statistics, it replaces the categorical features with numeric values. To do this, the CatBoost model generates multiple random permutations of the categorical feature. Then, the category k of feature x is replaced with an integer value calculated by the following formula:

$$avg_{target} = \frac{CountInClass + prior}{TotalCount + 1}, \quad (3)$$

where the variable *CountInClass* indicates the number of times the target value was equal to '1' for samples that have the current categorical feature value, the variable *TotalCount* shows the total number of samples up to the current one that have a categorical feature value matching the current one. The prior, p , could be found by taking the overall target average for the entire dataset. The reason for using different permutations of features is to reduce overfitting by introducing diversity into the model and making it less sensitive to the ordering of the categorical features. For more details on the mathematics of this model, please refer to the original paper by Liudmila Prokhorenkova et al. (2018).

3.2 | Voting ensemble methods

In this study, we build several ensemble models using the idea of soft voting ensemble models where for each sample, we extract the predicted probabilities of the two classes: bankrupt (shown by '1') and non-bankrupt (shown by '0') from the three models discussed in section 3.1. Subsequently, the final prediction for each specific sample is made by taking the average of these predictions. The unweighted averaging is simply taking the average over the outputs of all individual models. Assuming m classifiers (in our study, $m = 3$), where each individual classifier j predicts the probability of belonging to class 1 for observation i as Pr_{ij} . Consequently, the probability of belonging to class 0 is determined as $1 - Pr_{ij}$. As shown by formula (4), if the average of predicted class probabilities for observation i exceeds a threshold value (typically set as $c = 0.5$), then, observation i belongs to class 1, otherwise, it belongs to class 0.

$$\hat{y}_i = \begin{cases} 1 & \text{if } \frac{\sum_{j=1}^m Pr_{ij}}{m} \geq c \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The ensemble process that uses this method is biased towards weak baseline learners and is not a smart strategy for combining the baseline learners (Mohammed & Kora, 2023). The weighted averaging shown by formula (5) assigns different weights to the base learners based on their

predictive abilities. This approach aims to reduce the bias associated with the baseline models, ultimately leading to an improvement in the prediction performance.

$$\hat{y}_i = \begin{cases} 1 & \text{if } \frac{\sum_{j=1}^m w_j \cdot Pr_{ij}}{\sum_{j=1}^m w_j} \geq c \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

The choice of weights is an important aspect of the ensemble technique, as it determines how much influence each model has on the final prediction. In this study, we will implement different ways of weightings to find the highest prediction performance for our ensemble models.

3.2.1 | Ensemble 1: Unweighted or naive averaging

The first ensemble model is the unweighted averaging model, as explained earlier with formula (4). This ensemble model is considered as the base ensemble model, and our objective is to improve the results of this model by incorporating optimal weights into the averaging formula (5).

3.2.2 | Ensemble 2, 3: Weighted averaging with variable weighting

Figure 1 shows the overall framework of the ensemble models developed in this study. As indicated by formula (5), the weighted averaging model assigns different weights to individual classifiers in the calculation of final prediction probabilities. For our ensemble 2 and 3, we considered two sets of initial weights: [0.8, 0.1, and 0.1] and [0.5, 0.25, 0.25], respectively. While the choice of these initial weights was arbitrary, we aimed to have one set of weights with more similar weight (i.e., [0.5, 0.25, and 0.25]) versus another set that assigns a higher weight to one base classifier (i.e., weight 0.8). Through a 10-fold cross-validation, for each ensemble models 2 and 3, we assigned the highest weights in each set to the individual base learner that achieved the highest value for the AUC metrics in that specific fold. The rationale behind employing this metric is elaborated in Section 3.4. Then, the optimized weights, suitable for classifying out-of-sample data, were determined by averaging the assigned weights to

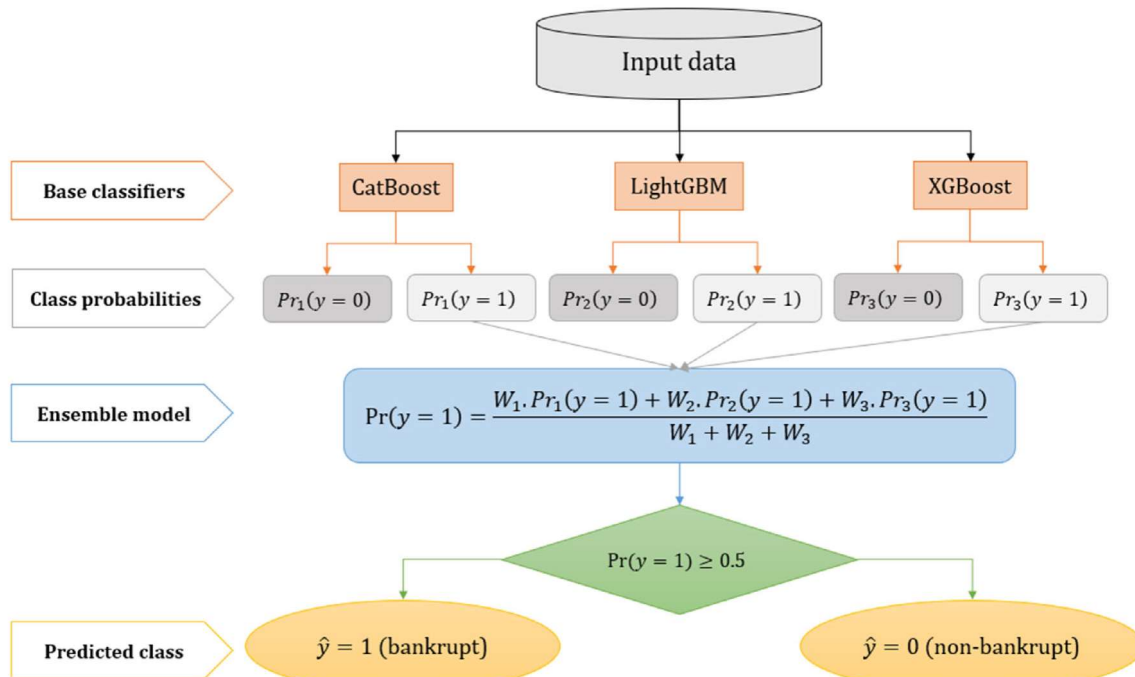


FIGURE 1 Overall framework of the proposed ensemble models.

each classifier over the 10 folds. For example, if 0.8 is assigned to XGBoost in six folds, then, its optimal weight will be calculated as $[(6)(0.8) + (4)(0.1)]/10$. The same interpretation applies to the other base classifiers in both ensemble models 2 and 3.

$$w_{CatBoost}^{(optimal)} = \frac{W_{CatBoost}^{(fold 1)} + \dots + W_{CatBoost}^{(fold 10)}}{10}, \quad (6)$$

$$w_{LightGBM}^{(optimal)} = \frac{W_{LightGBM}^{(fold 1)} + \dots + W_{LightGBM}^{(fold 10)}}{10}, \quad (7)$$

$$w_{XGBoost}^{(optimal)} = \frac{W_{XGBoost}^{(fold 1)} + \dots + W_{XGBoost}^{(fold 10)}}{10}. \quad (8)$$

3.2.3 | Ensemble 4: Weighted averaging with optimized weighting

Using the idea of super learner by van der Laan et al. (2007), we used a 10-fold cross-validation approach to find the optimal weights from predefined sets. The potential sets of weights considered in the optimization process are listed in Table 2. In each fold, all 27 distinct sets of weights were used in formula (5) to generate the weighted average of class probabilities. The optimal set is the one that achieves the highest average AUC value across the 10 folds.

3.3 | Data oversampling

To test whether the use of data oversampling techniques can improve our results, we will employ the Synthetic Minority Oversampling Technique (SMOTE) introduced by Chawla et al. (2002). This reason for employing this technique is that it has shown superiority over other techniques in similar studies such as Veganzones and Séverin (2018) and Garcia (2022). The basic idea behind SMOTE is that it identifies samples in the minority class that are in close proximity to each other. Subsequently, by randomly selecting a number of these samples, SMOTE interpolates a line between them. New samples are then generated along the line that connects the selected samples. The whole process is repeated until the desired ratio of minority to majority classes is achieved. It is worth mentioning that the SMOTE should be applied only to the training set and not the testing set. This is because the primary goal of SMOTE is to generate synthetic samples that helps the model see enough samples from the minority class so that it could learn the pattern and be generalized well on the real dataset with imbalanced situation.

3.4 | Hyperparameter tuning and evaluation metric

Following the works of other similar studies (e.g., Jabeur et al., 2021), we used the grid search method to find the optimal parameters for each base classifier. This method involves defining a grid of possible values for each hyperparameter. Subsequently, all combinations of the possible values are used in training the model. The combination that achieves the best prediction performance on the validation set is then selected as the optimal set of hyperparameters. The rational behind using the grid search method over the other methods like the random search is that the Grid search systematically evaluates every combination of hyperparameters specified in the grid, ensuring that no combination is missed. Random search selects hyperparameter values randomly from specified ranges. While randomness can help explore a wider range of hyperparameter space compared to grid search, it lacks the systematic exploration of specific combinations that grid search offers. This randomness may lead to

TABLE 2 The list of weights used in the optimization process of the ensemble 4.

$[w_{CatBoost}, w_{LightGBM}, w_{XGBoost}]$
[0.1, 0.1, 0.8] [0.1, 0.8, 0.1] [0.8, 0.1, 0.1]
[0.2, 0.2, 0.6] [0.2, 0.6, 0.2] [0.6, 0.2, 0.2]
[0.4, 0.2, 0.4] [0.4, 0.4, 0.2] [0.2, 0.4, 0.4]
[0.3, 0.2, 0.5] [0.2, 0.3, 0.5] [0.5, 0.3, 0.2] [0.5, 0.2, 0.3] [0.3, 0.5, 0.2] [0.2, 0.5, 0.3]
[0.1, 0.3, 0.6] [0.3, 0.1, 0.6] [0.6, 0.1, 0.3] [0.6, 0.3, 0.1] [0.1, 0.6, 0.3] [0.3, 0.6, 0.1]
[0.1, 0.2, 0.7] [0.2, 0.1, 0.7] [0.7, 0.1, 0.2] [0.7, 0.2, 0.1] [0.1, 0.7, 0.2] [0.2, 0.7, 0.1]

suboptimal hyperparameter settings. Random search results may be more difficult to reproduce compared to grid search. Since the selection of hyperparameter values is based on random sampling, reproducing the exact sequence of hyperparameter configurations evaluated during optimization may be challenging.

In prior research, Anggoro and Mukti (2021) compared the performance of grid search and random search methods for hyperparameter tuning in the Extreme Gradient Boosting (XGBoost) algorithm applied to predict chronic kidney failure. The grid search method could find the best hyperparameter combination and produced the best value, with an accuracy value of 99.28% and an f-measure of 0.9942. In another research, XGBoost has been applied alongside grid search tuning to predict dementia risk, achieving an accuracy of 85.61% (Ryu et al., 2020). Additionally, Zhang and Zhan (2017) demonstrated the effectiveness of employing grid search with XGBoost for classification tasks. Since we are employing the gradient Boosting methods, including the XGBoost, we take similar approach and employ the grid search method to find the best combination of hyperparameter values and maximize the performance of our models.

Table 3 shows the list of hyperparameters along with their possible values that we will use in our grid search optimization. We will employ a 10-fold cross-validation approach and using the AUC as the selection metric, we will find the optimal values of hyperparameters for each classifier, specifically for each classification case that will be explained in Section 4.1. More precisely, the set of hyperparameters that achieves the highest average AUC value across the 10 folds will be selected as the optimal set.

The reason for using the AUC criterion instead of the commonly used accuracy measure is the highly imbalanced nature of the dataset under consideration. In such situations, prediction models tend to classify all samples as the majority class, ignoring the minority class. Consequently, the accuracy of the model will be misleadingly high due to its good performance on classifying the majority class while potentially performing inadequately on the minority class. The AUC metric addresses this limitation by taking into account the accuracy of predictions for both classes, providing a more comprehensive evaluation of the model's performance across both classes. Most similar studies (e.g., Ben Jabeur & Serret, 2023; Veganzones & Séverin, 2018; Zoričák et al., 2020) utilize the AUC as their main performance metric to assess the prediction performance of their models.

4 | DATA AND RESULTS

4.1 | Data

Most studies rely on different financial ratios that are calculated from yearly financial statements (e.g., Garcia, 2022; Jabeur et al., 2021; Veganzones & Séverin, 2018; Zoričák et al., 2020) because these attributes give a view of firms' financial situation (Veganzones & Séverin, 2018). In this study, we use the financial data of polish companies provided by Zięba et al. (2016) which is available at this repository. According to Zięba et al. (2016), the period of time was established as 2007–2013 for bankrupt and 2000–2012 for still operating companies. In the period of 2007–2013 nearly 700 bankrupt enterprises were analysed. In the period 2000–2012 more than 10,000 still operating ones, (in this sample the company, which declared bankruptcy is excluded were taken into account) were taken into account. To create a single dataset that contains instances from two classes, Zięba et al. (2016) took samples from both bankrupt and non-bankrupted instances. Consequently, a dataset including 700 + 10,000 instances was created where each company has multiple financial statements over the period of 2000 to 2013 containing 64 financial indicators for each instance (See Table 4).

TABLE 3 List of hyperparameters for each base classifier.

CatBoost	LightGBM	XGBoost
depth = [3, 7, 12]	max_depth = [3, 7, 12]	max_depth = [5, 10]
learning_rate = [0.3, 0.1, 0.01]	learning_rate = [0.3, 0.1, 0.01]	learning_rate = [0.3, 0.1, 0.01]
iterations_set = [100, 500]	n_estimators = [100, 500]	n_estimators = [100, 500]
l2_leaf_reg = [1, 10]	reg_lambda = [1, 10]	gamma = [1, 10]
model_size_reg = [1, 10]	reg_alpha = [1, 10]	reg_lambda = [1, 10]
	min_split_gain = [0, 0.01, 0.1]	reg_alpha = [1, 10]
		min_child_weight = [1, 10]

Abbreviations: l2_leaf_reg, or reg_lambda; L2 regularization term on weights. It helps to prevent overfitting by penalizing large weights; Max_depth or depth, the maximum depth of the trees in the ensemble; min_child_weight, this hyperparameter helps control the growth of the tree and can be adjusted to avoid creating nodes with very few instances, which might be sensitive to noise and result in overfitting; min_split_gain, the minimum loss reduction required to make a further partition on a leaf node; model_size_reg, Regularization parameter for the model size. It controls the number of non-zero weights; n_estimators or iterations_set, the number of trees to be built in the ensemble. It represents the total number of boosting rounds; reg_alpha, L1 regularization term on weights. It induces sparsity by pushing some weights to exactly zero.

TABLE 4 The set of features considered in classification process taken from Zięba et al. (2016).

ID description	ID description
X1 net profit/total assets	X33 operating expenses/short-term liabilities
X2 total liabilities/total assets	X34 operating expenses/total liabilities
X3 working capital/total assets	X35 profit on sales/total assets
X4 current assets/short-term liabilities	X36 total sales/total assets
X5 [(cash + short-term securities + receivables – short-term liabilities)/(operating expenses – depreciation)] * 365, X37 (current assets – inventories)/long-term liabilities	X38 constant capital/total assets
X6 retained earnings/total assets	X39 profit on sales/sales
X7 EBIT/total assets	X40 (current assets – inventory – receivables)/short-term liabilities
X8 book value of equity/total liabilities	X41 total liabilities/((profit on operating activities + depreciation) * (12/365))
X9 sales/total assets	X42 profit on operating activities/sales
X10 equity/total assets	X43 rotation receivables + inventory turnover in days
X11 (gross profit + extraordinary items + financial expenses)/total assets	X44 (receivables * 365)/sales
X12 gross profit/short-term liabilities	X45 net profit/inventory
X13 (gross profit + depreciation)/sales	X46 (current assets – inventory)/short-term liabilities
X14 (gross profit + interest)/total assets	X47 (inventory * 365)/cost of products sold
X15 (total liabilities * 365)/(gross profit + depreciation)	X48 EBITDA (profit on operating activities – depreciation)/total assets
X16 (gross profit + depreciation)/total liabilities	X49 EBITDA (profit on operating activities – depreciation)/sales
X17 total assets/total liabilities	X50 current assets/total liabilities
X18 gross profit/total assets	X51 short-term liabilities/total assets
X19 gross profit/sales	X52 (short-term liabilities * 365)/cost of products sold
X20 (inventory * 365)/sales	X53 equity/fixed assets
X21 sales (n)/sales (n-1)	X54 constant capital/fixed assets
X22 profit on operating activities/total assets	X55 working capital
X23 net profit/sales	X56 (sales – cost of products sold)/sales
X24 gross profit (in 3 years)/total assets	X57 (current assets – inventory – short-term liabilities)/(sales – gross profit – depreciation)
X25 (equity – share capital)/total assets	X58 total costs /total sales
X26 (net profit + depreciation)/total liabilities	X59 long-term liabilities/equity
X27 profit on operating activities/financial expenses	X60 sales/inventory
X28 working capital/fixed assets	X61 sales/receivables
X29 logarithm of total assets	X62 (short-term liabilities * 365)/sales
X30 (total liabilities – cash)/sales	X63 sales/short-term liabilities
X31 (gross profit + interest)/sales	X64 sales/fixed assets
X32 (current liabilities * 365)/cost of products sold	

- For companies labelled as bankrupt in this dataset, their bankruptcy status was determined in a year between 2007 and 2013, and they have at least one financial report in the 5 years leading to bankruptcy.
- For companies labelled as non-bankrupt, their financial statements cover the years 2000 to 2012, and they have at least three consecutive financial statements in this period.

Since one of our objectives is to improve the results of Zięba et al. (2016), we use the same classification cases or periods that were defined by Zięba et al. (2016) as below:

- Year 1: data from 1st year of the forecasting period and corresponding class label that indicates bankruptcy status after 5 years.
- Year 2: data from 2nd year of the forecasting period and corresponding class label that indicates bankruptcy status after 4 years.
- Year 3: data from 3rd year of the forecasting period and corresponding class label that indicates bankruptcy status after 3 years.
- Year 4: data from 4th year of the forecasting period and corresponding class label that indicates bankruptcy status after 2 years.
- Year 5: data from 5th year of the forecasting period and corresponding class label that indicates bankruptcy status after 1 year.

Table 5 shows the class distributions in each classification case. As can be seen, we are dealing with a highly imbalanced dataset in which there are more observations with class 0 (i.e., non-bankrupt companies) than the observations with class 1 (i.e., bankrupt companies).

TABLE 5 Class distributions in each of the five classification periods.

Year	N	% class 1 (bankrupt)	% class 0 (non-bankrupt)
year1	7027	3.86	96.14
year2	10,173	3.93	96.07
year3	10,503	4.71	95.29
year4	9792	5.26	94.74
year5	5910	6.94	93.06

As preprocessing steps, missing data were filled in with the mean values following the work of Du et al. (2020). We did not remove any attribute from the dataset as we wanted to keep the same dataset that was utilized by Zięba et al. (2016).

4.2 | Experimental protocol

For the data oversampling experiments, we employed the SMOTE technique to generate two sets of extra samples for each year: one with bankrupt to non-bankrupt ratio of 0.5, and the other with ratio of 1:1, ensuring an equal number of bankrupt and non-bankrupt samples in the whole dataset.

We designed two sets of experiments: Similar to the study framework of Papík and Papíková (2023) and Zięba et al. (2016), we divided the entire dataset into 10 folds, and we validated our three base classifiers using the cross-validation technique. The optimal weights for the ensemble models were also determined in this step. Then, to show the effectiveness of our optimized ensemble models, we randomly divided the entire dataset into two parts where we used 90% of the data for training the models with their optimal parameters and used the remaining 10% for evaluating their prediction performance. To reduce the impact of randomness, we repeated this data splitting 10 times with 10 different random numbers and reported the average values of the AUC on ten hold-out test sets.

In case of cross-validation, the SMOTE was applied to the nine folds comprising the training set, while the remaining fold which was acting as the test set, remained untouched to resemble the true distribution of classes in the real world. Similarly, for the case of hold-out test sets, we exclusively applied the SMOTE technique to the training sets. Since the SMOTE method generates synthetic samples, it increases the size of the training set. To avoid this and preserve the original size of the training sets without oversampling, we randomly removed the extra samples from the majority class, specifically the existing non-bankrupt samples in the training sets.

4.3 | Results

We conducted experiments for all five classification cases described in Section 4.1 and under different oversampling scenarios, namely using the ratios of 0.5 and 1. The results are presented in panel b, c, and d of Table 6. Since one of our objectives in this study was to improve the results of Zięba et al. (2016), we included their top three sets of results corresponding to XGB (XGBoost) and two extensions of this model in panel a of Table 6. Moreover, Table 7 shows the optimal weights of the ensemble models determined through the 10-fold cross-validation strategy.

Looking at the results from the 10-fold cross-validation shown in Table 6, it is evident that nearly all ensemble models outperform their base learners (indicated in bold). This is true regardless of using the original or oversampled data. Comparing the results reported in panel b and a, the AUC values obtained for the original data (without oversampling) outperform the best results reported by Zięba et al. (2016). In addition, the standard deviations of our results are smaller than those reported in panel a, indicating the increased robustness of our models in comparison to the top models of Zięba et al. (2016).

When we look at the AUC values achieved by the models on hold-out-test sets, we can see that the ensemble models with optimal weights outperform their base learners on a hold-out test set as well. This means that optimal weights are reliable enough to be utilized in our ensemble models to make prediction for unseen (new) data.

Comparing the results reported in panels b, c, and d, it is clear that for most of the years (classification cases), the data oversampling technique degrades the performance of our models. This indicates that the data oversampling technique that has been widely used by researchers as a treatment for imbalanced data classification, does not increase the performance of gradient boosted models such as XGBoost, LightGBM, and CatBoost. However, this observation is specific to the SMOTE method and is limited to our dataset. We do not generalize this to other data sampling methods, such as undersampling of the majority class or alternative versions of SMOTE employed by other researchers.

By looking at the optimal weights represented in Table 7, we notice that each year is associated with a dominant classifier, accounting for the highest share of predictions in the ensemble models for that particular year. For example, the dominant classifier in year 1 is the LightGBM model

TABLE 6 AUC achieved by each model and for each prediction year under different oversampling scenarios.

Panel a: Benchmark results from Zięba et al. (2016)												
Year1		Year2		Year3		Year4		Year5				
10-fold CV												
Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	
XGB	0.945	0.330	0.917	0.027	0.922	0.025	0.935	0.024	0.951	0.024		
XGBE	0.953	0.024	0.941	0.019	0.929	0.049	0.94	0.027	0.954	0.018		
EXGB	0.959	0.018	0.944	0.021	0.940	0.032	0.941	0.025	0.955	0.019		
Panel b: The original data with no oversampling												
Year1		Year2		Year3		Year4		Year5		Year1		
10-fold CV										Hold-out test set		
	Std. Dev		Std. Dev		Std. Dev		Std. Dev		Std. Dev		Std. Dev	
Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	
CatBoost	0.961	0.017	0.941	0.021	0.941	0.019	0.947	0.013	0.961	0.015	0.958	
LightGBM	0.961	0.015	0.932	0.023	0.939	0.022	0.947	0.013	0.958	0.017	0.960	
XGBoost	0.952	0.019	0.938	0.020	0.937	0.019	0.948	0.010	0.959	0.015	0.956	
Ensemble 1	0.964	0.015	0.944	0.021	0.944	0.019	0.953	0.011	0.963	0.014	0.964	
Ensemble 2	0.967	0.015	0.948	0.021	0.946	0.019	0.954	0.010	0.965	0.015	0.964	
Ensemble 3	0.966	0.015	0.946	0.021	0.945	0.019	0.954	0.010	0.964	0.015	0.964	
Ensemble 4	0.965	0.016	0.945	0.021	0.944	0.019	0.953	0.010	0.963	0.014	0.963	
Panel c: Oversampled data with a ratio of 0.5 for bankrupt to non-bankrupt companies												
Year1		Year2		Year3		Year4		Year5		Year1		
10-fold CV										Hold-out test set		
	Std. Dev		Std. Dev		Std. Dev		Std. Dev		Std. Dev		Std. Dev	
Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	
CatBoost	0.951	0.018	0.927	0.020	0.929	0.021	0.942	0.023	0.954	0.009	0.949	
LightGBM	0.954	0.020	0.921	0.019	0.926	0.023	0.938	0.017	0.953	0.007	0.948	
XGBoost	0.943	0.022	0.927	0.020	0.929	0.025	0.937	0.021	0.953	0.010	0.948	
Ensemble 1	0.953	0.018	0.93	0.018	0.933	0.023	0.943	0.020	0.956	0.007	0.954	
Ensemble 2	0.956	0.017	0.932	0.017	0.935	0.023	0.946	0.020	0.958	0.007	0.953	
Ensemble 3	0.954	0.017	0.931	0.018	0.934	0.023	0.945	0.020	0.957	0.007	0.954	
Ensemble 4	0.955	0.018	0.931	0.019	0.933	0.022	0.944	0.021	0.957	0.007	0.952	
												0.016
(Continues)												

(Continues)

TABLE 6 (Continued)

	Year1		Year2		Year3		Year4		Year5		Year1		Year2		Year3		Year4		Year5	
	10-fold CV										Hold-out test set									
	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev	Mean	Std. Dev
CatBoost	0.947	0.029	0.922	0.025	0.928	0.022	0.934	0.013	0.952	0.017	0.953	0.019	0.921	0.021	0.927	0.011	0.936	0.014	0.947	0.010
LightGBM	0.955	0.028	0.910	0.026	0.923	0.020	0.927	0.016	0.950	0.016	0.957	0.017	0.915	0.014	0.922	0.017	0.931	0.014	0.946	0.012
XGBoost	0.948	0.028	0.916	0.02	0.921	0.022	0.929	0.014	0.950	0.017	0.955	0.019	0.908	0.022	0.921	0.011	0.934	0.013	0.945	0.011
Ensemble ₁	0.955	0.026	0.920	0.023	0.929	0.020	0.934	0.013	0.953	0.016	0.959	0.018	0.921	0.017	0.931	0.010	0.938	0.012	0.948	0.010
Ensemble ₂	0.957	0.026	0.924	0.023	0.930	0.021	0.937	0.012	0.954	0.016	0.959	0.018	0.923	0.017	0.932	0.009	0.939	0.011	0.948	0.010
Ensemble ₃	0.956	0.026	0.922	0.023	0.930	0.020	0.935	0.013	0.954	0.016	0.959	0.018	0.921	0.017	0.932	0.010	0.939	0.011	0.948	0.010
Ensemble ₄	0.956	0.026	0.924	0.025	0.930	0.021	0.935	0.012	0.953	0.016	0.959	0.017	0.924	0.017	0.932	0.009	0.939	0.011	0.949	0.010

TABLE 7 Optimal weights for the ensemble models suitable for using in out-of-sample classification.

Model	The original data with no oversampling				
	Year1	Year2	Year3	Year4	Year5
Ensemble 2 optimal weights	[0.38, 0.45 , 0.17]	[0.45 , 0.17, 0.38]	[0.52 , 0.31, 0.17]	[0.31, 0.24, 0.45]	[0.52 , 0.24, 0.24]
Ensemble 3 optimal weights	[0.35, 0.38 , 0.28]	[0.38 , 0.28, 0.35]	[0.4 , 0.32, 0.28]	[0.32, 0.3, 0.38]	[0.4 , 0.3, 0.3]
Ensemble 4 optimal weights	[0.6 , 0.3, 0.1]	[0.6 , 0.1, 0.3]	[0.5 , 0.3, 0.2]	[0.4, 0.2, 0.4]	[0.5 , 0.2, 0.3]
	Oversampled data with a ratio of 0.5 for bankrupt to non-bankrupt companies				
	Year1	Year2	Year3	Year4	Year5
Ensemble 2 optimal weights	[0.31, 0.59 , 0.1]	[0.38, 0.24, 0.38]	[0.38, 0.24, 0.38]	[0.52 , 0.24, 0.24]	[0.38 , 0.31, 0.31]
Ensemble 3 optimal weights	[0.32, 0.42 , 0.25]	[0.35, 0.3, 0.35]	[0.35, 0.3, 0.35]	[0.4 , 0.3, 0.3]	[0.35 , 0.32, 0.32]
Ensemble 4 optimal weights	[0.2, 0.7 , 0.1]	[0.6 , 0.1, 0.3]	[0.5 , 0.2, 0.3]	[0.6 , 0.2, 0.2]	[0.6 , 0.1, 0.3]
	Oversampled data with a ratio of 1:1 for bankrupt to non-bankrupt companies				
	Year1	Year2	Year3	Year4	Year5
Ensemble 2 optimal weights	[0.24, 0.52 , 0.24]	[0.52 , 0.24, 0.24]	[0.59 , 0.17, 0.24]	[0.59 , 0.17, 0.24]	[0.38 , 0.31, 0.31]
Ensemble 3 optimal weights	[0.3, 0.4 , 0.3]	[0.4 , 0.3, 0.3]	[0.42 , 0.28, 0.3]	[0.42 , 0.28, 0.3]	[0.35 , 0.32, 0.32]
Ensemble 4 optimal weights	[0.3, 0.6 , 0.1]	[0.8 , 0.1, 0.1]	[0.6 , 0.2, 0.2]	[0.7 , 0.1, 0.2]	[0.6 , 0.2, 0.2]

as it has the largest weight across various ensemble models and different data oversampling scenarios. Conversely, the dominant classifier in years 2, 3, and 5 is the CatBoost. In year 4, XGBoost claims the dominance when there is no data oversampling, while CatBoost prevails in scenarios involving data oversampling. This suggests that CatBoost exhibits greater resilience to changes of data, leading to more accurate predictions in the presence of data oversampling. Overall, the CatBoost is the most reliable and accurate classifier, holding the highest weight in 30 out of 45 sets of optimal weights presented in Table 7.

5 | DISCUSSION

To assess the robustness of our analysis, we employed the Wilcoxon signed-rank test (Wilcoxon, 1945) to test the statistical significance of differences between the results obtained by each model and those obtained by all other models. This method is particularly useful in situations where the data do not meet the assumptions of parametric tests such as the t-test. Similar studies (Almaskati et al., 2021; Zięba et al., 2016) have employed this test to assess whether there are statistically significant differences in the predictive accuracy between the various models being evaluated.

In both the 10-fold cross-validation and hold-out test set scenarios, and across all classification periods and models, we generated 10 AUC values. In the cross-validation step, these values correspond to the AUC scores of the 10 folds. In the hold-out test case, we obtained 10 AUC values from randomly splitting the entire dataset into training and testing sets, resulting in 10 hold-out test sets. Therefore, for each pair of models, we can test the following hypotheses:

H_0 : The median of the differences between the AUC values of Model 1 and Model 2 is zero

H_a : The median of the differences between the AUC values of Model 1 and Model 2 is positive

If H_0 is rejected, that means there is a better than 50% chance that the AUC values of Model 1 are greater than those of Model 2. The p -values presented in Tables 8–10 confirm the previous conclusions drawn in Sections 4.3. First, across all years, almost all ensemble models consistently outperform their base learners, regardless of whether original or oversampled data is used. This is evident from the predominance of green colour in the bottom half of the tables, where the p -values comparing ensemble models to base classifiers are significantly small to reject the null hypothesis at a significance level of 10% or even 5%. This means that the medians of the differences between AUC values of ensemble models and base classifiers are positive.

The same interpretation can be made for the hold-out test set scenario. Therefore, we can conclude that the optimized weights are suitable for use in ensemble models for predicting unseen data, given the high performance of ensemble models compared to their base learners. Since the p -values comparing ensemble 2 to other models are less than 10% for almost all cases, we can conclude that it is the best ensemble model in

our study. Depending on the classification case (year), we can utilize the best set of weights from Table 7 for this model and make prediction for new out-of-sample data.

Regarding the general economic conditions over the period of our dataset, according to the article titled ‘Poland’s Economic Performance Amid Global Financial Crisis’¹ by the economist Witold M. Orłowski, Poland recorded impressive growth figures, demonstrating continuous economic expansion despite the global financial crisis in 2009. Several factors have been discussed in this article, collectively indicate a stable economic environment in Poland during the period of our study (2000–2013). In light of this information, we posit that the economic conditions in Poland remained relatively unchanged and resilient throughout our study period. Therefore, our analysis focuses on predicting bankruptcy outcomes within the context of a stable economic backdrop.

Implementing ensemble models has its own challenges and difficulties. Selecting diverse and complementary models within an ensemble to increase overall performance is a challenge. Achieving this diversity while ensuring the models complement each other is a non-trivial task and it demands a deep understanding of the characteristics of the dataset and the underlying learning problem. While CatBoost, XGBoost, and LightGBM are all gradient boosting algorithms, they have different underlying implementations and optimization techniques. Each algorithm may capture different patterns in the data due to its unique learning strategies. Differences in default hyperparameter configurations, even for common parameters, can lead to variations in how each model interacts with the dataset. For instance, learning rates, tree depths, and regularization strengths may differ. The errors made by each model on the dataset can differ. Instances where one model fails but others succeed contribute to ensemble diversity. The diversity introduced by CatBoost, XGBoost, and LightGBM can collectively enhance the ensemble’s ability to generalize well and make accurate predictions across various scenarios.

Tuning hyperparameters for different models within an ensemble model is complex. The challenge is figuring out the best combinations of hyperparameters for each individual model to boost its performance and, at the same time, ensuring that these models work well together to improve the overall effectiveness of the ensemble. In this study, we tuned hyperparameters of each model individually since Individual tuning facilitates a deep understanding of how each algorithm responds to hyperparameter adjustments, promoting interpretability. In addition, focusing on individual tuning aims to maximize the standalone performance of each base learner, leading to higher performances when their outputs are used in the ensemble model. Tuning all models together may risk overfitting the ensemble to the training data, especially if there is a large hyperparameter search space. Most importantly, simultaneous tuning can be computationally intensive, especially when exploring a wide range of hyperparameter combinations for each model.

We trained the base learners sequentially at each fold which means a step-by-step training process, where each model is trained one after another. Our approach is efficient for a smaller number of models and folds, but as the size of ensemble models or the complexity of the dataset increases, parallel training can significantly reduce overall training time by simultaneously processing multiple models. In general, if computational resources are limited or if parallelization is not cost-effective, sequential training may be more practical.

The user-intervention level in implementing ensemble models refers to the degree of manual input, domain knowledge, and subjective choices made by the user throughout the ensemble modelling process. In our implementation, users may manually define the initial weights used in ensemble 2, 3, and 4. All other processes like the hyperparameter tuning, weights optimization and the outputs’ aggregation are done automatically. This can be considered as “low Intervention”.

6 | CONCLUSION

This study designs several heterogeneous ensemble models to predict bankruptcy status of companies in a highly imbalanced dataset which resembles the real world. Using gradient boosting classifiers as the base learners for our ensemble models, we showed that not only the ensemble models have significantly better prediction performance compared to their base learners, but also their enhancement of the best results from a benchmark paper using the same dataset.

We also showed that the widely adopted data oversampling technique, often considered a remedy for imbalanced data issues, deteriorates the predictive ability of our models. This could have different reasons such as the nature of our data which is a high dimensional dataset, or the inherent characteristics of gradient boosting algorithms which might perform better on the original data. We are interested in exploring the possible reasons as an extension of this study.

Our experimental framework comprises two key steps: In our 10-fold cross-validation strategy, we showed the effectiveness of our ensemble models in the bankruptcy prediction problem. Using the optimal weights derived from the cross-validation step, we proved that our optimized ensemble models can be used in making accurate predictions on hold-out test sets which can be generalized to unseen or out-of-sample data. Our conclusions hold consistent across all prediction periods and under various data oversampling cases with different ratios of bankrupt to non-bankrupt companies.

Our study has several implications for managers, banks, financial institutions, investors, and more. By using an accurate prediction model, managers can predict whether their company will go bankrupt after 1, 2, or even 5 years, enabling them to avoid that by taking early preventive actions. Banks and financial institutions can rely on accurate predictions to assess their clients’ creditworthiness which help them in their decision

TABLE 8 Wilcoxon signed-rank test results—The original data without oversampling.

10-fold CV							Hold-out test set						
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
CatBoost	LightGBM	0.577	0.042	0.385	0.539	0.161	CatBoost	LightGBM	0.754	0.014	0.053	0.652	0.080
CatBoost	XGBoost	0.097	0.348	0.161	0.652	0.188	CatBoost	XGBoost	0.348	0.019	0.116	0.500	0.348
CatBoost	Ensemble1	0.920	0.784	0.903	0.968	0.884	CatBoost	Ensemble1	0.947	0.976	0.884	0.990	0.813
CatBoost	Ensemble2	0.995	0.981	0.999	0.995	0.947	CatBoost	Ensemble2	0.968	0.981	0.947	0.993	0.935
CatBoost	Ensemble3	0.968	0.903	0.990	0.986	0.935	CatBoost	Ensemble3	0.947	0.976	0.884	0.993	0.884
CatBoost	Ensemble4	0.981	0.903	0.990	0.968	0.935	CatBoost	Ensemble4	0.981	0.993	0.947	0.995	0.903
LightGBM	CatBoost	0.461	0.968	0.652	0.500	0.862	LightGBM	CatBoost	0.278	0.990	0.958	0.385	0.935
LightGBM	XGBoost	0.014	0.862	0.161	0.754	0.784	LightGBM	XGBoost	0.116	0.423	0.652	0.216	0.968
LightGBM	Ensemble1	0.884	1.000	0.968	0.990	0.997	LightGBM	Ensemble1	0.986	1.000	1.000	1.000	0.998
LightGBM	Ensemble2	0.976	1.000	0.990	0.997	0.997	LightGBM	Ensemble2	0.986	1.000	1.000	1.000	0.997
LightGBM	Ensemble3	0.947	1.000	0.976	0.997	0.997	LightGBM	Ensemble3	0.986	1.000	1.000	1.000	0.997
LightGBM	Ensemble4	0.903	0.998	0.968	0.990	0.986	LightGBM	Ensemble4	0.722	1.000	1.000	1.000	0.997
XGBoost	CatBoost	0.920	0.688	0.862	0.385	0.839	XGBoost	CatBoost	0.688	0.986	0.903	0.539	0.688
XGBoost	LightGBM	0.990	0.161	0.862	0.278	0.246	XGBoost	LightGBM	0.903	0.615	0.385	0.813	0.042
XGBoost	Ensemble1	0.998	0.903	0.997	0.986	0.998	XGBoost	Ensemble1	0.990	1.000	0.997	0.998	0.993
XGBoost	Ensemble2	0.999	0.990	0.999	0.997	0.999	XGBoost	Ensemble2	0.981	1.000	0.990	1.000	0.981
XGBoost	Ensemble3	0.998	0.947	0.999	0.990	1.000	XGBoost	Ensemble3	0.981	1.000	0.995	0.999	0.990
XGBoost	Ensemble4	0.995	0.935	0.992	0.986	0.995	XGBoost	Ensemble4	0.935	1.000	0.993	0.996	0.990
Ensemble1	CatBoost	0.097	0.246	0.116	0.042	0.138	Ensemble1	CatBoost	0.065	0.032	0.138	0.014	0.216
Ensemble1	LightGBM	0.138	0.001	0.042	0.014	0.005	Ensemble1	LightGBM	0.019	0.001	0.001	0.001	0.003
Ensemble1	XGBoost	0.003	0.116	0.005	0.019	0.003	Ensemble1	XGBoost	0.014	0.001	0.005	0.003	0.010
Ensemble1	Ensemble2	1.000	1.000	1.000	0.884	0.975	Ensemble1	Ensemble2	0.615	0.976	0.839	0.065	0.722
Ensemble1	Ensemble3	1.000	1.000	1.000	0.999	0.999	Ensemble1	Ensemble3	0.839	0.981	0.839	0.080	0.652
Ensemble1	Ensemble4	0.862	0.754	0.839	0.461	0.688	Ensemble1	Ensemble4	0.246	0.968	0.843	0.246	0.722
Ensemble2	CatBoost	0.007	0.024	0.002	0.007	0.065	Ensemble2	CatBoost	0.042	0.024	0.065	0.010	0.080
Ensemble2	LightGBM	0.032	0.001	0.014	0.005	0.005	Ensemble2	LightGBM	0.019	0.001	0.001	0.001	0.005
Ensemble2	XGBoost	0.002	0.014	0.002	0.005	0.002	Ensemble2	XGBoost	0.024	0.001	0.014	0.001	0.024
Ensemble2	Ensemble1	0.001	0.001	0.001	0.138	0.025	Ensemble2	Ensemble1	0.423	0.032	0.188	0.947	0.313
Ensemble2	Ensemble3	0.024	0.002	0.001	0.278	0.216	Ensemble2	Ensemble3	0.500	0.065	0.246	0.947	0.461
Ensemble2	Ensemble4	0.024	0.002	0.007	0.097	0.097	Ensemble2	Ensemble4	0.246	0.615	0.429	0.784	0.423
Ensemble3	CatBoost	0.042	0.116	0.014	0.019	0.080	Ensemble3	CatBoost	0.065	0.032	0.138	0.010	0.138
Ensemble3	LightGBM	0.065	0.001	0.032	0.005	0.005	Ensemble3	LightGBM	0.019	0.001	0.001	0.001	0.005

(Continues)

TABLE 8 (Continued)

10-fold CV		Hold-out test set											
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
Ensemble3	XGBoost	0.003	0.065	0.002	0.014	0.001	Ensemble3	XGBoost	0.024	0.001	0.007	0.002	0.014
Ensemble3	Ensemble1	0.001	0.001	0.001	0.002	0.002	Ensemble3	Ensemble1	0.188	0.024	0.188	0.935	0.385
Ensemble3	Ensemble2	0.981	0.999	1.000	0.754	0.813	Ensemble3	Ensemble2	0.539	0.947	0.784	0.065	0.577
Ensemble3	Ensemble4	0.246	0.348	0.042	0.024	0.339	Ensemble3	Ensemble4	0.216	0.958	0.862	0.524	0.476
Ensemble4	CatBoost	0.024	0.116	0.014	0.042	0.080	Ensemble4	CatBoost	0.024	0.010	0.065	0.007	0.116
Ensemble4	LightGBM	0.116	0.003	0.042	0.014	0.019	Ensemble4	LightGBM	0.313	0.001	0.001	0.001	0.005
Ensemble4	XGBoost	0.007	0.080	0.008	0.019	0.007	Ensemble4	XGBoost	0.080	0.001	0.010	0.004	0.014
Ensemble4	Ensemble1	0.161	0.278	0.188	0.577	0.348	Ensemble4	Ensemble1	0.784	0.042	0.157	0.784	0.313
Ensemble4	Ensemble2	0.981	0.999	0.995	0.920	0.920	Ensemble4	Ensemble2	0.784	0.423	0.571	0.246	0.615
Ensemble4	Ensemble3	0.784	0.688	0.968	0.981	0.661	Ensemble4	Ensemble3	0.813	0.053	0.161	0.476	0.524

TABLE 9 Wilcoxon signed-rank test results—Oversampled data with a ratio of 0.5 for bankrupt to non-bankrupt companies.

10-fold CV							Hold-out test set						
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
CatBoost	LightGBM	0.784	0.116	0.216	0.138	0.385	CatBoost	LightGBM	0.423	0.042	0.032	0.097	0.161
CatBoost	XGBoost	0.010	0.577	0.539	0.116	0.461	CatBoost	XGBoost	0.278	0.019	0.246	0.348	0.348
CatBoost	Ensemble1	0.903	0.903	0.920	0.652	0.920	CatBoost	Ensemble1	0.920	0.722	0.722	0.935	0.920
CatBoost	Ensemble2	0.995	0.990	0.995	0.935	0.976	CatBoost	Ensemble2	0.920	0.754	0.839	0.981	0.935
CatBoost	Ensemble3	0.976	0.947	0.958	0.884	0.968	CatBoost	Ensemble3	0.903	0.784	0.813	0.958	0.947
CatBoost	Ensemble4	0.981	0.986	0.968	0.903	0.968	CatBoost	Ensemble4	0.862	0.903	0.839	0.986	0.976
LightGBM	CatBoost	0.246	0.903	0.813	0.884	0.652	LightGBM	CatBoost	0.615	0.968	0.976	0.920	0.862
LightGBM	XGBoost	0.014	0.981	0.903	0.246	0.539	LightGBM	XGBoost	0.652	0.754	0.615	0.976	0.813
LightGBM	Ensemble1	0.116	0.999	0.997	0.998	1.000	LightGBM	Ensemble1	0.999	0.995	0.997	1.000	0.999
LightGBM	Ensemble2	0.384	1.000	0.998	0.999	1.000	LightGBM	Ensemble2	1.000	0.997	0.995	0.999	0.998
LightGBM	Ensemble3	0.161	0.999	0.997	0.998	1.000	LightGBM	Ensemble3	1.000	0.995	0.997	1.000	0.999
LightGBM	Ensemble4	0.348	0.993	0.993	0.995	0.986	LightGBM	Ensemble4	0.999	0.993	0.993	0.998	0.993
XGBoost	CatBoost	0.993	0.461	0.500	0.903	0.577	XGBoost	CatBoost	0.754	0.986	0.784	0.688	0.688
XGBoost	LightGBM	0.990	0.024	0.116	0.784	0.500	XGBoost	LightGBM	0.385	0.278	0.423	0.032	0.216
XGBoost	Ensemble1	1.000	0.998	0.968	1.000	0.958	XGBoost	Ensemble1	0.997	1.000	0.976	0.958	0.990
XGBoost	Ensemble2	1.000	1.000	0.997	1.000	0.995	XGBoost	Ensemble2	0.986	1.000	0.995	0.947	0.995
XGBoost	Ensemble3	1.000	1.000	0.981	1.000	0.993	XGBoost	Ensemble3	0.995	1.000	0.981	0.958	0.995
XGBoost	Ensemble4	1.000	0.995	0.968	1.000	0.968	XGBoost	Ensemble4	0.986	1.000	0.990	0.920	0.981
Ensemble1	CatBoost	0.116	0.116	0.097	0.385	0.097	Ensemble1	CatBoost	0.097	0.313	0.313	0.080	0.097
Ensemble1	LightGBM	0.903	0.002	0.005	0.003	0.001	Ensemble1	LightGBM	0.002	0.005	0.005	0.001	0.002
Ensemble1	XGBoost	0.001	0.003	0.042	0.001	0.053	Ensemble1	XGBoost	0.007	0.001	0.032	0.053	0.014
Ensemble1	Ensemble2	0.995	1.000	0.997	0.947	0.990	Ensemble1	Ensemble2	0.188	0.968	0.813	0.862	0.975
Ensemble1	Ensemble3	1.000	1.000	0.998	0.997	0.993	Ensemble1	Ensemble3	0.688	0.813	0.813	0.754	0.994
Ensemble1	Ensemble4	0.976	0.862	0.754	0.784	0.839	Ensemble1	Ensemble4	0.053	0.935	0.976	0.754	0.615
Ensemble2	CatBoost	0.007	0.014	0.007	0.080	0.032	Ensemble2	CatBoost	0.097	0.278	0.188	0.024	0.080
Ensemble2	LightGBM	0.616	0.001	0.003	0.002	0.001	Ensemble2	LightGBM	0.001	0.005	0.007	0.002	0.003
Ensemble2	XGBoost	0.001	0.001	0.005	0.001	0.007	Ensemble2	XGBoost	0.019	0.001	0.007	0.065	0.007
Ensemble2	Ensemble1	0.007	0.001	0.005	0.065	0.014	Ensemble2	Ensemble1	0.839	0.053	0.216	0.161	0.025
Ensemble2	Ensemble3	0.010	0.080	0.010	0.161	0.010	Ensemble2	Ensemble3	0.903	0.032	0.313	0.339	0.278
Ensemble2	Ensemble4	0.042	0.313	0.007	0.053	0.007	Ensemble2	Ensemble4	0.014	0.884	0.952	0.163	0.577
Ensemble3	CatBoost	0.032	0.065	0.053	0.138	0.042	Ensemble3	CatBoost	0.116	0.246	0.216	0.053	0.080
Ensemble3	LightGBM	0.862	0.002	0.005	0.003	0.001	Ensemble3	LightGBM	0.001	0.005	0.005	0.001	0.002

(Continues)

TABLE 9 (Continued)

10-fold CV		Hold-out test set											
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
Ensemble3	XGBoost	0.001	0.001	0.024	0.001	0.010	Ensemble3	XGBoost	0.007	0.001	0.024	0.053	0.007
Ensemble3	Ensemble1	0.001	0.001	0.003	0.005	0.010	Ensemble3	Ensemble1	0.348	0.187	0.187	0.278	0.006
Ensemble3	Ensemble2	0.993	0.935	0.993	0.862	0.993	Ensemble3	Ensemble2	0.116	0.976	0.722	0.661	0.784
Ensemble3	Ensemble4	0.920	0.722	0.116	0.385	0.246	Ensemble3	Ensemble4	0.024	0.920	0.968	0.348	0.524
Ensemble4	CatBoost	0.024	0.019	0.042	0.116	0.042	Ensemble4	CatBoost	0.161	0.116	0.188	0.019	0.032
Ensemble4	LightGBM	0.688	0.010	0.010	0.007	0.019	Ensemble4	LightGBM	0.002	0.010	0.010	0.003	0.010
Ensemble4	XGBoost	0.001	0.007	0.042	0.001	0.042	Ensemble4	XGBoost	0.019	0.001	0.014	0.097	0.024
Ensemble4	Ensemble1	0.032	0.161	0.278	0.246	0.188	Ensemble4	Ensemble1	0.958	0.080	0.032	0.313	0.423
Ensemble4	Ensemble2	0.968	0.722	0.995	0.958	0.995	Ensemble4	Ensemble2	0.990	0.138	0.048	0.837	0.461
Ensemble4	Ensemble3	0.097	0.313	0.903	0.652	0.784	Ensemble4	Ensemble3	0.981	0.097	0.042	0.722	0.476

TABLE 10 Wilcoxon signed-rank test results—Oversampled data with a ratio of 1:1 for bankrupt to non-bankrupt companies.

10-fold CV		Hold-out test set											
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
CatBoost	LightGBM	0.993	0.005	0.032	0.024	0.188	CatBoost	LightGBM	0.935	0.246	0.278	0.080	0.423
CatBoost	XGBoost	0.784	0.024	0.014	0.097	0.246	CatBoost	XGBoost	0.862	0.024	0.065	0.188	0.476
CatBoost	Ensemble1	0.998	0.161	0.862	0.539	0.813	CatBoost	Ensemble1	0.976	0.539	0.935	0.839	0.862
CatBoost	Ensemble2	1.000	0.976	0.995	0.920	0.976	CatBoost	Ensemble2	0.968	0.688	0.986	0.986	0.862
CatBoost	Ensemble3	0.998	0.688	0.976	0.688	0.903	CatBoost	Ensemble3	0.976	0.577	0.981	0.920	0.862
CatBoost	Ensemble4	1.000	0.968	0.976	0.839	0.920	CatBoost	Ensemble4	0.968	0.884	0.993	0.993	0.903
LightGBM	CatBoost	0.010	0.997	0.976	0.981	0.839	LightGBM	CatBoost	0.080	0.784	0.754	0.935	0.615
LightGBM	XGBoost	0.010	0.976	0.138	0.903	0.500	LightGBM	XGBoost	0.348	0.138	0.461	0.976	0.500
LightGBM	Ensemble1	0.423	0.999	1.000	1.000	0.920	LightGBM	Ensemble1	0.958	0.995	0.986	1.000	0.968
LightGBM	Ensemble2	0.958	1.000	1.000	1.000	0.976	LightGBM	Ensemble2	0.986	0.998	0.986	1.000	0.958
LightGBM	Ensemble3	0.652	0.999	1.000	1.000	0.957	LightGBM	Ensemble3	0.968	0.997	0.986	1.000	0.968
LightGBM	Ensemble4	0.862	1.000	0.999	0.999	0.935	LightGBM	Ensemble4	0.990	0.997	0.990	0.999	0.947
XGBoost	CatBoost	0.246	0.981	0.990	0.920	0.784	XGBoost	CatBoost	0.161	0.981	0.947	0.839	0.524
XGBoost	LightGBM	0.993	0.032	0.884	0.116	0.539	XGBoost	LightGBM	0.688	0.884	0.577	0.032	0.539
XGBoost	Ensemble1	1.000	0.981	0.999	0.993	0.997	XGBoost	Ensemble1	0.993	1.000	0.999	0.999	0.958
XGBoost	Ensemble2	1.000	0.999	1.000	0.999	0.999	XGBoost	Ensemble2	0.986	0.999	1.000	0.993	0.957
XGBoost	Ensemble3	1.000	0.993	0.999	0.997	0.997	XGBoost	Ensemble3	0.986	0.999	1.000	0.998	0.958
XGBoost	Ensemble4	0.999	0.997	0.997	0.986	0.981	XGBoost	Ensemble4	0.968	0.997	1.000	0.990	0.947
Ensemble1	CatBoost	0.003	0.862	0.161	0.500	0.216	Ensemble1	CatBoost	0.032	0.500	0.080	0.188	0.161
Ensemble1	LightGBM	0.615	0.002	0.001	0.001	0.097	Ensemble1	LightGBM	0.053	0.007	0.019	0.001	0.042
Ensemble1	XGBoost	0.001	0.024	0.002	0.010	0.005	Ensemble1	XGBoost	0.010	0.001	0.002	0.002	0.053
Ensemble1	Ensemble2	0.995	0.995	0.981	0.999	0.999	Ensemble1	Ensemble2	0.385	0.993	0.813	0.862	0.884
Ensemble1	Ensemble3	0.999	0.999	0.999	0.996	0.986	Ensemble1	Ensemble3	0.813	0.990	0.993	0.947	0.528
Ensemble1	Ensemble4	0.986	0.997	0.843	0.935	0.688	Ensemble1	Ensemble4	0.577	0.903	0.903	0.784	0.722
Ensemble2	CatBoost	0.001	0.032	0.007	0.097	0.032	Ensemble2	CatBoost	0.042	0.348	0.019	0.019	0.161
Ensemble2	LightGBM	0.053	0.001	0.001	0.001	0.032	Ensemble2	LightGBM	0.019	0.003	0.019	0.001	0.053
Ensemble2	XGBoost	0.001	0.002	0.001	0.002	0.002	Ensemble2	XGBoost	0.019	0.002	0.001	0.010	0.043
Ensemble2	Ensemble1	0.007	0.007	0.024	0.002	0.002	Ensemble2	Ensemble1	0.688	0.010	0.246	0.188	0.138
Ensemble2	Ensemble3	0.042	0.010	0.297	0.007	0.065	Ensemble2	Ensemble3	0.884	0.007	0.461	0.348	0.138
Ensemble2	Ensemble4	0.116	0.500	0.097	0.042	0.019	Ensemble2	Ensemble4	0.813	0.839	0.862	0.539	0.652
Ensemble3	CatBoost	0.003	0.348	0.032	0.348	0.116	Ensemble3	CatBoost	0.032	0.461	0.024	0.097	0.161
Ensemble3	LightGBM	0.385	0.002	0.001	0.001	0.043	Ensemble3	LightGBM	0.042	0.005	0.019	0.001	0.042

(Continues)

TABLE 10 (Continued)

10-fold CV		Hold-out test set											
Model1	Model2	Year1	Year2	Year3	Year4	Year5	Model1	Model2	Year1	Year2	Year3	Year4	Year5
Ensemble3	XGBoost	0.001	0.010	0.002	0.005	0.005	Ensemble3	XGBoost	0.019	0.002	0.001	0.003	0.053
Ensemble3	Ensemble1	0.002	0.002	0.002	0.004	0.019	Ensemble3	Ensemble1	0.216	0.014	0.014	0.065	0.472
Ensemble3	Ensemble2	0.968	0.993	0.703	0.995	0.947	Ensemble3	Ensemble2	0.161	0.995	0.577	0.688	0.903
Ensemble3	Ensemble4	0.958	0.990	0.423	0.539	0.313	Ensemble3	Ensemble4	0.500	0.862	0.779	0.652	0.722
Ensemble4	CatBoost	0.001	0.042	0.032	0.188	0.097	Ensemble4	CatBoost	0.042	0.138	0.010	0.010	0.116
Ensemble4	LightGBM	0.161	0.001	0.002	0.002	0.080	Ensemble4	LightGBM	0.014	0.005	0.014	0.002	0.065
Ensemble4	XGBoost	0.002	0.005	0.005	0.019	0.019	Ensemble4	XGBoost	0.042	0.005	0.001	0.014	0.065
Ensemble4	Ensemble1	0.019	0.005	0.157	0.080	0.348	Ensemble4	Ensemble1	0.500	0.116	0.116	0.246	0.313
Ensemble4	Ensemble2	0.903	0.500	0.920	0.968	0.986	Ensemble4	Ensemble2	0.216	0.188	0.161	0.500	0.423
Ensemble4	Ensemble3	0.053	0.014	0.652	0.500	0.722	Ensemble4	Ensemble3	0.539	0.161	0.221	0.385	0.313

making on how to allocate the loans which results in better risk and resource management strategies. It also helps investors to make better decisions and avoid financial losses. It could be used as a guide for practitioners and academics in designing bankruptcy prediction models for other datasets with different settings.

For the future studies, we are interested in using and comparing various deep learning models as our base learners of more complex ensemble models. Furthermore, we plan to explore the application of our approach on more recent datasets to validate the robustness of our findings. Particularly, during the last 4 years, the rate of bankruptcy has been increased due to the global crisis caused by the COVID-19 pandemic.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in UCI Machine Learning Repository at <https://doi.org/10.24432/C5F600>.

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ENDNOTE

¹ <https://blogs.lse.ac.uk/eurompblog/2012/09/25/poland-economic-success-orlowski/>.

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